ONLINE RESEOURCE 2

ELECTRONIC SUPPLEMENTARY MATERIAL (ESM-2)

INFLAMMATION RESEARCH

The role of CD8+ T lymphocytes in chronic obstructive pulmonary disease: a systematic review.

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Table S1: Studies investigating the presence of CD8+ T lymphocytes in COPD. Ten human studies investigated the number of CD8+ T lymphocytes in COPD.

Publication	Title	Subjects	COPD diagnosis	Sample type	Conclusions
Baraldo et al [1] Human 2015	Immune activation in a1 anti-trypsin deficiency emphysema (AATD). Beyond the protease-antiprotease paradigm	10 AATD, 26 COPD, 17 control smokers (S), 25 healthy non-smokers (HNS)	AATD – serum levels and genotyping COPD – macroscopic quantification of emphysema, method of Heard	Explanted lung samples	Those with usual COPD had a significantly greater number of CD8+ T cells than S and HNS Those with AATD had significantly greater number of CD8+ T cells than HNS Suggests AATD is similar to usual COPD
Eapen et al [2] Human 2017	Profiling cellular and inflammatory changes in the airway wall of mild to moderate COPD	Large airway: 13 COPD- current smokers (CS), 14 COPD-ex smokers (exS), 20 S, 25 HNS Small airway: 9 COPD-CS, 10 COPD exS, 11 S, 10 HNS	COPD – Global Initiative for Obstructive Lung Diseases (GOLD) criteria COPD – GOLD stage I/II	Endobronchial biopsy Lung tissue	Large airway: CD8+ T cells were the dominant lymphocyte CD8+ T cells were lower than normal in smoker/COPD groups but there was a change in the CD8+/CD4+ T cell ratio. In COPD-CS there was a marked relative excess of CD8+ over CD4+ Small airway: CD8+ T cells were higher than normal across all clinical groups and especially COPD-CS
Siena et al [3] Human 2011	Reduced apoptosis of CD8+ T-lymphocytes in the airways of smokers with mild/ moderate COPD	All COPD smokers: 5 mild, 11 moderate, 3 severe, 6 very severe COPD 16 S, 16 HNS	COPD – GOLD classification I-IV	Lung tissue specimens	The median percentage of apoptotic CD8+ T cells in the submucosa of both the central and peripheral airways was significantly lower in mild/moderate COPD than in HNS, S and severe/very severe COPD When all subjects are considered, no correlation exists between the apoptosis of CD8+ T cells and functional parameters but when severe/very severe COPD are excluded, there is a significant positive correlation between the percentage of apoptotic CD8+ T cells and FEV_1 and FEV_1/FVC in both the central and peripheral airways A significant negative correlation was determined between total CD8+ T lymphocytes and the percentage of apoptotic CD8+ T cells in central and peripheral airways

Lofdahl et al [4] Human 2008	Increased intraepithelial T-cells in stable COPD	22 COPD, 14 S, 15 HNS	COPD – GOLD	Endobronchial mucosal biopsy sample	The number of intraepithelial CD8+ T cells per millimetre of epithelium was significantly higher in COPD compared to HNS, but no significant difference was determined when compared to S
Chen et al [5] Human 2012	Imbalance of circulating T-lymphocyte subpopulation in COPD and its relationship with CAT performance	All COPD: 13 moderate, 14 severe, 25 very severe	American Thoracic Society/ European Respiratory Society (ATS/ERS) standards COPD Assessment Test (CAT score)	Peripheral blood	Proportion of CD8+ T cells in the group with CAT>30 was significantly higher than other groups. There was no significant difference in the proportion of CD8+ T cells in groups with 20<score£30, 10<score£20 and score£10 Positive correlation between the proportion of CD8+ T cells and CAT score when score>20 but not when score£20 No difference in the proportion of CD8+ T cells in the peripheral blood between moderate, severe, extremely severe COPD when comparing by GOLD stage
Mathai et al [6] Human 2013	Peripheral blood T-cell populations in COPD, asymptomatic smokers and healthy non-smokers in Indian subpopulation – a pilot study	21 COPD, 19 S, 20 HNS	History of cough with sputum and/or dyspnoea on most days of the month for at least 3 months for at least 2 consecutive years, and airflow limitation $FEV_1/FVC < 0.70$ and $FEV_1 < 80\%$ predicted	Peripheral blood	There was no difference in the absolute counts of peripheral blood T cell subsets among HNS, S and COPD There was no difference between the mean CD8+ T cell counts when comparing COPD, S and HNS No correlation was determined between CD8+ T cells and smoking intensity, or with the type of smoking (bidi or cigarette)

Forsslund et al [7] Human 2014	Distribution of T-cell subsets in BAL fluid of patients with mild to moderate COPD depends on current smoking status and not airway obstruction	27 COPD-CS, 11 COPD-exS, 40 S, 40 HNS	GOLD stage I-II	Bronchoalveolar lavage fluid (BAL) Peripheral blood	The percentage of CD8+ T cells was significantly higher in BAL from COPD- CS and S There was a lower percentage of CD8+ T cells in BAL from COPD-exS compared to both COPD-CS and S. The percentage of CD8+ T cells in BAL from COPD-CS was positively correlated with the number of cigarettes smoked per day in the last 6 months. This correlation was even stronger in male COPD-CS In the peripheral blood, COPD-CS and S groups had a lower median percentage of CD8+ T cells than HNS.
Olloquequi et al [8] Human 2010	Differential lymphocyte infiltration in small airways and lung parenchyma in COPD patients	16 moderate COPD, 16 very severe COPD, 18 S, 9 HNS	GOLD stage II (moderate) or GOLD stage IV (very severe)	Excised lungs or lobes Peripheral blood	Small airway: COPD had increased density of CD8+ T cells than controls. Subjects with moderate COPD had a significantly higher density of CD8+ T cells in the epithelial layer, and also in the total small airway than HNS. there were no differences in CD8+ T cell density in smooth muscle. Lung parenchyma: the CD8+ T cell density was significantly higher in the interstitium of COPD compared to controls. There was no significant difference in CD8+ T cell density in areas proximal and distal to emphysema in COPD lungs. Moderate COPD and S groups had significantly higher CD8+ T cells in the bronchial wall compared to parenchymal interstitium The number of pack years smoked was positively correlated with CD8+ T cell density in the total small airway, bronchiolar connective compartment and epithelium, and the number of CD8+ T cells in the alveolar interstitium Peripheral blood – no difference in cell numbers among 4 clinical groups

Urboniene et al [9] Human 2013	Distribution of gd and other T-lymphocyte subsets in patients with chronic obstructive pulmonary disease and asthma	20 COPD, 18 asthma, 14 healthy subjects	GOLD stage II and III	Induced sputum BAL Peripheral blood	The percentage and absolute numbers of CD8+ T cells in induced sputum was significantly higher in COPD compared to asthmatics and healthy subjects. CD8+ T cells were also higher in BAL in COPD compared to asthmatics and healthy subjects. CD8+ T cell number in both BAL and induced sputum correlated negatively with FEV_1 There was no difference in the number of CD8+ T cells in peripheral blood between any study groups.
Hodge et al [10] Human 2007	Increased intracellular T helper 1 pro-inflammatory cytokine production in peripheral blood, bronchoalveolar lavage and intraepithelial T cells of COPD subjects	7 COPD-CS, 5 COPD-CS- using inhaled corticosteroids (ics) , 7 COPD-exS, 6 COPD-exS- ics, 7 S, 8 HNS	GOLD	BAL Bronchial mucosa Peripheral blood	BAL: percentage of CD8+ T cells was significantly increased in COPD-CCS and S compared to HNS. The percentage of CD8+ T cells producing IFN-g and TNF-a was increased in COPD and S compared to HNS. COPD-CS also had a significantly higher percentage of CD8+ T cells producing IFN-g and TNF-a than COPD-exS Bronchial mucosa: the percentage of CD8+ T cells was increased in COPD- exS-ics and S compared to HNS. A significantly higher proportion of CD8+ T cells producing TNF-a in all COPD and S compared to HNS. The percentage of CD8+ T cells producing IFN-g and TNF-a was significantly lower in COPD-CS-ics than COPD-CS Peripheral blood: there was a ssignificantly higher number of CD8+ T cells in COPD-CS and COPD-exS compared to HNS. The proportion of CD8+ T cells producing IFN-g and TNF-a was significantly higher in COPD-CS and COPD-exS than S and HNS